FieldForge: The Virtual Physics World

Complete Research & Strategy Document

Prepared: 27 September 2026

Subject: A component-based, deterministically-verified 3D physics simulation world — problem in, lawful simulation out, no AI at runtime — with an MCP interface so AI agents can build on it.

1. Executive Summary

The idea in one paragraph. FieldForge is a virtual world that owns a complete library of physics components (resistors, batteries, wires, lenses, projectiles…), every formula physics has ever produced, the laws and principles those components obey, and the tools used to measure them — all stored as vast, rigorously interlinked tables. When a student inputs a problem, the system reads off the required components, which collapses the universe of physics down to a small relevant subset; it then resolves what must be calculated by recursively chasing missing values through the links until everything is known; finally it assembles the selected components in 3D and animates them in the sequence the solution was derived — showing not just the answer, but *how things connect and work*. No AI model runs at solve time. Every number is computed through a verified formula; every animation frame is backed by a law. Separately, an MCP (Model Context Protocol) server exposes the world to AI agents, turning every AI tutor from a competitor into a customer of FieldForge's ground truth.

The headline verdict. This is a genuinely original combination with a real, documented gap in the market. Nothing today generates a verified 3D physics simulation live from a problem statement with a sequenced component build-up. The moat is not the idea — it is the *curated, machine-checked knowledge base*, which compounds with every chapter encoded. **Overall rating: 7.4/10** — strong concept and timing, with execution weight (content curation + 3D quality bar) as the central risk. Full scoring and critique in §4.

Market context. The virtual/remote laboratories market stood at ~USD 1.65B in 2025, forecast to ~USD 5.35B by 2034 (~14% CAGR); interactive STEM simulation ~USD 3.0B (2026) → ~USD 10.4B (2034, ~16.8% CAGR). The free tier is dominated by PhET (150+ sims, 20+ years, free) and Falstad (still the most-used free circuit simulator). The paid tier: Labster (2,000+ institutions), ExploreLearning Gizmos (550+ sims), Brilliant (~$130/yr), Wolfram Alpha Pro ($5–12/mo), and enterprise tools (COMSOL, Ansys). AI-generated simulations are the emerging threat — and the opening: published builders admit AI sims can *teach wrong physics with the authority of interactive visualization* (e.g., implementing $1/r^3$ instead of $1/r^2$). FieldForge's "lawful, not plausible" positioning is the direct antidote.

2. The Idea, Distilled

2.1 The problem it solves

Students anywhere in the world have no access to a physics component library — no shop of parts, no lab bench, no way to *see* physics working. Textbooks show static diagrams; videos show someone else's experiment; AI chatbots describe answers in words. None of them let a student grab a battery, wire it to a resistor, and watch current, fields, and forces emerge — with every quantity justified by a law.

2.2 The mechanism (the funnel)

- **Problem in.** "A 10Ω resistor in series with a 5V battery…" — the problem statement already contains the parts list.
- **Grab components.** Battery, resistor, wires are pulled from the shop. Each component arrives with its inbuilt characteristic properties (a wire has resistance; two parallel wires have a magnetic field between them).
- **Collapse the universe.** The vast tables (all formulas, all laws) are joined down through the selected components to a small relevant subset — a database join, not a search.
- **Resolve the target.** What must be calculated (current)? Find formulas for current among the subset; if an input is missing, chase its links — which component holds it, which formula derives it — recursively, until everything is known.
- **Assemble and animate.** The components are arranged per the problem; the resolution order becomes the animation sequence: connect battery → current flows → field generated → force appears. The solver's trace *is* the storyboard.
- **Show all paths.** Where multiple valid derivations exist, show every path — the shortest and the alternatives. AI tutors trained on public solutions show the one canonical path; a world that derives from components finds all of them.

2.3 The two products in one

- **For humans:** the virtual lab bench — grab, wire, watch, probe, learn.
- **For AI agents:** an MCP server exposing the component world — agents get physics ground truth instead of hallucinating simulations. Every AI tutor becomes a distribution channel, not a competitor.

2.4 Design principles

- **No AI at runtime.** Zero token cost, zero API keys, works offline, fully ownable. AI may assist *authoring* the knowledge base offline; it never decides physics at solve time.
- **Lawful, not plausible.** Every number computed through a verified formula; every frame backed by a law. Plausibility is the enemy.

- **Components are the index.** You never search "all of physics" — components collapse the search space by construction.
- **The trace is the storyboard.** Derivation order determines animation order; no separate animation planner needed.

3. Architecture Fundamentals: Linking the Tables So Nothing Is Left Unconnected

This is the heart of the engineering question: *how do formulas, components, properties, laws, principles, and tools all connect such that no content is orphaned and no problem falls through a crack?* The answer has five parts: a canonical data model, linking contracts, completeness rules, a build strategy for the huge tables, and the resolution algorithm.

3.1 The canonical data model — the property is the universal key

Everything links through **properties**, not through components or formulas directly. A property is a physical quantity with a canonical identity:

```
property {
  key:        "electric.current"        # globally unique, dotted namespace
  symbol:     "I"
  dimension:  [0, 0, -1, 1, 0, 0, 0]   # SI exponents: kg m s A K mol cd
  unit_family: ["A", "mA", "uA", ...]
  domain:     "electromagnetism"
}
```

Why this matters: "current" means the same thing whether it appears in a resistor's behavior, a formula, or a wire's field. The property key is the foreign key of the entire system. Two tables never link to each other directly — they link *through* shared property keys. This single decision is what makes cross-topic reasoning (projectile + electric field, circuits + magnetics) fall out naturally instead of requiring special cases.

3.2 The tables and their linking contracts

```
Table  |  What it holds  |  Linking contract
components  |  Every physical component: resistor, battery, wire, capacitor, lens, projectile…  |  Declares the property keys it *carries*, each with a role: `intrinsic` (fixed by nature, e.g., resistivity), `settable` (user assigns, e.g., resistance=10Ω), `observable` (emerges from solving, e.g., current through it)
formulas  |  Every physics formula  |  Declares `solves_for: <property key>` and `needs: [<property keys>]`, plus a dimension-checked expression in a safe DSL
laws  |  Conservation laws, Maxwell's equations, Newton's laws…  |  Declares the *scope* of properties/components it governs (e.g., KCL governs every circuit node; energy conservation
```

```
governs isolated systems)
principles  |  Superposition, symmetry arguments, idealizations  |  Declares applicability conditions as property predicates
tools  |  Ammeter, voltmeter, oscilloscope, ruler…  |  Modeled *as components* — a measurement is just placing another component in the world. No special-case "probe" system
units  |  Unit families per dimension  |  Every property key maps to exactly one dimension vector; conversions are mechanical
```

The discipline: **a formula never names a component; a component never names a formula.** Both name only property keys. This is what keeps the graph clean as it scales to tens of thousands of rows — no pairwise spaghetti, only hub-and-spoke through properties.

3.3 The completeness rules — the "nothing left" guarantee

"Nothing is left unconnected" is not a hope; it is four mechanically checkable rules run as continuous integration over the tables:

- **No orphan property.** Every property key must be *solvable by ≥1 formula* OR *settable on ≥1 component* OR *derivable from ≥1 law*. A property nobody can produce is a dead end — the audit flags it and curation fixes it.
- **No dangling reference.** Every `needs` and `solves_for` in every formula must resolve to an existing property key. Every component's carried properties must exist. Broken on insert — the compiler rejects it the way a type-checker rejects bad code.
- **Dimension closure.** Every formula's expression must dimensionally balance (left-hand dimension vector equals right-hand). Checked at authoring time, not at solve time. Invalid physics cannot enter the building.
- **Coverage closure.** A seed set of curriculum problems (e.g., every end-of-chapter problem in the target textbook chapter) must *resolve* — the resolver must find at least one complete path for each. This runs as CI. Coverage is a number on a dashboard, not a feeling.

These four rules convert "huge tables" from a liability into an asset: the bigger the tables get, the more the audit guarantees they stay coherent. **The knowledge base is a compiled artifact, and the compiler is the dimension-checker plus the audit.**

3.4 Building the huge tables — staged bootstrapping

Do not attempt "all of physics" on day one. The build order:

- **Phase 0 — one domain, one chapter.** DC circuits, Class 11–12 / JEE current-electricity: ~40 components, ~120 formulas, full coverage of that chapter's problem set. A complete small world beats an incomplete infinite one.
- **Phase 1 — chapter by chapter.** Each new chapter is a curation sprint: encode components → encode formulas with source citations (formulas are public-domain knowledge; textbooks are the source) → run the coverage CI → fix orphans the audit finds.

The audit *drives* the backlog: "properties with zero solving formulas" is a to-do list.

- **Phase 2 — cross-domain links.** Mechanics (particles, fields) comes second; the property-hub design means linking a charged projectile to an electric field requires no schema change — just new rows.
- **Phase 3 — community curation.** A formula workbench (DSL editor + live dimension check + "test against problem set") lets contributors add knowledge the compiler verifies. Wikipedia's model, but every edit is machine-checked.

Estimated seed effort for Phase 0: one domain expert (this is where the founder's electrical engineering background is the unfair advantage) can encode DC circuits in weeks, not months — the knowledge is finite and well-organized in any textbook.

3.5 The resolution algorithm — the funnel, precisely

Given: component instances (from the problem) with set values, and a target property.

```
1. COLLECT  properties touched by the selected components      # the join
2. CANDIDATES = formulas where solves_for == target
                and formula.properties ⊆ collected            # confined subset
3. For each candidate (shortest-first):
     RESOLVE(inputs):
       for each input property p:
         if p has a set value → ok
         elif ∃ formula f solving p with resolvable inputs → RESOLVE(f.needs)
         elif p settable on a present component → prompt/assume
         else → fail this branch (try next candidate)
     Constraints: cycle detection (visited-set on property keys),
                  depth limit, memoization of resolved subtrees
4. EVALUATE the winning resolution tree bottom-up through the DSL
5. ENUMERATE all complete trees → the alternative paths
6. TRACE → ordered list of (component, property, formula, value) → the storyboard
```

Complexity is tamed by construction: step 1 collapses the formula space from "all of physics" to "formulas touching these components" (typically dozens, not thousands) before any search begins. This is why the architecture beats a naive formula-graph search — **the components are an index, and indexes exist to make search unnecessary.**

3.6 Where the simulation lives

The resolution trace compiles directly into the simulation spec:

- **Scene:** the component instances, arranged per the problem's topology (series/parallel/wiring captured at input).
- **Timeline:** each trace step becomes an animation beat — component appears, value resolves, quantity visualizes (current dots flow, field arrows grow, force vectors draw).
- **Live data:** time-dependent quantities are sampled across t to drive motion and charts.

- **Probes:** tools are components, so "measure the current" is placing an ammeter — its reading is just another resolved property.

The 3D renderer is deliberately downstream and dumb: it renders whatever the spec says. All the intelligence is in the tables and the resolver. This separation is what lets visual quality scale independently of physics correctness.

4. Critical Review & Rating

4.1 Scorecard

```
Dimension | Score | Notes
Novelty | 8/10 | The *combination* — component-indexed resolution + lawful animation + MCP exposure — has no direct equivalent. Pieces exist separately (see §5).
Market timing | 8/10 | AI-era trust crisis in generated content favors "verified/lawful"; MCP ecosystem exploding (22,000+ servers catalogued); STEM sim market growing 14–17%/yr.
Technical feasibility | 7/10 | DC circuits fully tractable (nodal analysis is textbook); cross-domain particle/field solving is harder; the real grind is content curation, not algorithms.
Defensibility (day 1) | 6/10 | The architecture is copyable by a funded team. The moat is execution: coverage depth + curation velocity + user trust. Rises to 8/10 with a 2-year lead.
Business viability | 7/10 | Multiple monetizable surfaces (B2C subscription, B2B institutional, B2D API/MCP usage). Edtech buyers exist and pay.
Execution risk | 4/10 | The red flag: huge content surface × high 3D quality bar × single founder. This is where projects like this die — not on ideas, on throughput.
Overall | 7.4/10 | A genuinely original, well-timed idea whose success depends on ruthless scoping and curation velocity.
```

4.2 The five hardest criticisms (steelmanned)

- **"All formulas that ever existed" is a trap.** An unbounded knowledge goal guarantees permanent incompleteness. The fix is curricular scoping: define "complete" as "every problem in textbook X, chapters Y–Z resolves." Ship complete small worlds; expand chapter by chapter. Completeness must be a *testable* property (§3.3, rule 4), never an aspiration.

- **The "discovers new physics" claim will get you laughed out of the room if overstated.** The system discovers new *paths*, not new *laws* — laws are inputs. Frame it as *combinatorial creativity within fixed rules* (the AlphaGo analogy: novel moves, same rules). The genuinely exciting version is cross-topic paths no textbook shows (the cavity-projectile chain), which is defensible and demonstrable.

- **Text → components is the hardest AI-free step.** Extracting *topology* ("in series

with a parallel combination of…") from prose without ML is genuinely difficult. Mitigation, in order: (a) builder-first UI — text supplies the parts list, hands do the wiring (Tinkercad proved this UX); (b) constrained templates for common topologies; (c) deterministic parsing only for values/units (the easy part). Do not promise "paste any problem."

- **"Top notch" 3D is the most expensive promise.** Visual excellence is a grinding, asset-heavy discipline. Falstad conquered circuit simulation with 2D dots. Start schematic, earn the right to beautiful — and note the architecture *supports* this: the renderer is downstream of the spec (§3.6), so visual upgrades never touch the physics.

- **Cold start: an empty shop is worthless.** The product has zero value until one domain is fully stocked. This argues for the narrowest possible MVP (DC circuits, one chapter, fully covered) and against any breadth before depth. The completeness audit (§3.3) is the tool that tells you when a domain is "done."

4.3 What the idea gets right (the genuine insights)

- **Components as an index into knowledge** is the single best architectural decision: it converts search into joins and makes the system faster as knowledge grows, not slower.
- **The trace-is-the-storyboard** insight eliminates an entire subsystem (no separate animation planner) and guarantees every visual is justified — the core of the "lawful" claim.
- **Tools-as-components** dissolves the probe/measurement special case elegantly.
- **The MCP inversion** (agents as customers, not competitors) is strategically the most valuable idea in the document — see §6.
- **No-AI-at-runtime** is not just cost-saving; it is a *trust* position in an era of hallucinated content, and a *reliability* position (offline, deterministic, ownable).

5. Market Landscape

5.1 Free / open-source competitors

```
Product | What it is | Cost | Strength vs FieldForge | Gap FieldForge exploits
PhET (Univ. of Colorado) | 150+ research-backed interactive sims, physics/chem/math | Free, partly open-source | 20+ years of pedagogy research; massive teacher trust; translated widely | Every sim hand-built (months each); no problem-to-sim generation; 2D
Falstad / CircuitJS | Browser circuit simulator with animated current flow | Free, open-source | Still the most-used free circuit sim (2025); instant, no signup | 2D schematic only; user must already know what to build; no derivation/explanation
oPhysics / MyPhysicsLab | HTML5 physics interactives | Free | Lightweight, embeddable | Single-concept toys; no composition across topics
OpenModelica | Industrial component-based multi-physics modeling | Free, open-source | The closest architectural ancestor (acausal components!) | Engineer-targeted; brutal learning curve; not educational; no 3D problem-to-sim
```

```
LTspice / ngspice / Qucs | Professional SPICE circuit simulators | Free | Gold-standard accuracy for circuits | Professional EDA UX; zero pedagogy; no animation narrative
Amrita Virtual Labs (India, govt.) | Government-funded virtual labs | Free | Curriculum-aligned for Indian syllabus | Scripted experiments, not generative; dated UX
```

5.2 Paid competitors

```
Product | What it is | Price signal | Strength | Gap
Labster | Gamified virtual labs, 2,000+ institutions (MIT, Stanford…) | Institutional (raised $34.7M+; now ~50 competitors in space) | High fidelity, curriculum-aligned, LMS-integrated | Expensive; content is hand-authored scenarios; no generative problem solving
ExploreLearning Gizmos | 550+ STEM sims, grades 3–12 | Subscription (school licenses) | Huge library, standards-aligned | 2D, fixed scenarios; can't compose novel setups
Brilliant | Interactive STEM courses | ~$130–150/yr premium | Best-in-class interactive pedagogy; strong brand | No real simulation bench; guided paths only, no open lab
Wolfram Alpha Pro | Computational answers + steps | $5–12/mo | Exact computation; enormous knowledge base | Answers, not worlds; no 3D; query must be precise; closed
COMSOL / Ansys | Enterprise multiphysics simulation | $$$$ (thousands/yr) | Ultimate fidelity | Professional-only; no education UX; massive cost
```

5.3 The AI-native threat (and why it validates the idea)

Educators already build simulations with Claude Artifacts — Anthropic's own education research reports teachers creating "custom simulation, interactive experiment" in hours, where PhET needed months per sim. **This is the threat and the validation simultaneously.** The threat: infinite bespoke sims at zero marginal cost. The opening, documented by builders themselves: *"A simulation that looks correct but implements $1/r^3$ instead of $1/r^2$ is worse than no simulation at all. It's teaching wrong physics with the authority of interactive visualization."* (Vivek Karmarkar, Jan 2026). AI sims are **plausible**; FieldForge is **lawful**. In education — where wrong physics taught authoritatively is actively harmful — verifiability is not a feature, it is *the* feature. Every AI-generated sim is a future customer of a verification layer.

5.4 Market sizing

- **Virtual & remote laboratories:** ~USD 1.65B (2025) → ~USD 5.35B (2034), ~14.0% CAGR. Platform/software is 56% of revenue and gaining share.
- **Interactive STEM simulation:** ~USD 3.0B (2026) → ~USD 10.4B (2034), ~16.8% CAGR. Drivers: lab access limits, budget constraints, safety rules, unlimited repetition at zero consumable cost.
- **Simulation learning (broad):** ~USD 24.4B (2025) → ~USD 53.5B (2030), ~16–19% CAGR.
- **Geography:** Asia-Pacific has the highest forecast CAGR — India/China investing heavily in virtual labs for massive student populations. The founder's home market is the

growth market.

5.5 The gap map — where FieldForge sits

```
                        hand-authored                    generative
                              │                               │
   2D schematic ──────────────┼───────────────────────────────┤
   Falstad, oPhysics          │ PhET, Gizmos                  │  ← nobody here
                              │                               │
   3D / immersive ────────────┼───────────────────────────────┤
   (enterprise only)          │ Labster                       │  ← FIELD-FORGE:
                              │                               │  generative + lawful
   verified ──────────────────┴───────────────────────────────┘  + 3D + sequenced
   (laws checked)                                                + multi-path
```

No competitor occupies **generative × lawful × 3D × sequenced-build × multi-path**. The nearest neighbors each miss at least two: PhET (not generative), Falstad (not 3D/explanatory), Labster (not generative), Claude-artifacts (not lawful), OpenModelica (not educational/generative-UX).

5.6 What you can get for free vs. what you must build

- **Free:** formula knowledge itself (public domain — encode from textbooks); SPICE-class solvers (ngspice source as reference); three.js for 3D; Postgres; the MCP spec and SDKs.
- **Paid/expensive to build:** the curated, linked, machine-checked tables (this is the moat — labor, not licenses); high-quality 3D component models; curriculum alignment work; institutional sales motion.
- **Strategic note:** nothing on the free list is differentiating. Everything differentiating is on the build list — which is good news, because it means the moat is *earned*, not bought.

6. The MCP Play: A Physics World for AI Agents

6.1 The inversion

Today's AI tutors (ChatGPT, Claude, Khanmigo-style agents) *describe* physics in words and occasionally generate *unverified* simulations. They cannot guarantee a single number. FieldForge's MCP server turns this around: **agents don't compete with the world — they operate inside it.** The agent handles language and pedagogy; FieldForge handles truth.

6.2 Proposed MCP tool surface

```
Tool  |  What the agent can do
```

```
`list_components` / `get_component`  |  Browse the shop; read a component's properties, laws, 3D model
`assemble`  |  Create a world: place components, set values, define topology
`connect`  |  Wire components (series/parallel/topology)
`solve`  |  Run the resolver; get all solution paths with values
`probe`  |  Place a tool-component (ammeter…) and read a quantity
`derive`  |  Return the human-readable derivation (all paths) for tutoring
`render_frames`  |  Get the sequenced animation spec for the solution trace
```

Every tool returns *verified* results — the agent can narrate freely, but it cannot invent physics, because it never computes physics. **The world is the source of truth; the agent is the storyteller.**

6.3 Why this is strategically the best idea in the document

- **Distribution:** every AI tutor, homework helper, and edtech chatbot becomes a potential integrator. You don't out-market OpenAI/Anthropic — you *power* their physics.
- **Ecosystem timing:** 22,000+ MCP servers are now catalogued; registries, install flows, and agent tooling are maturing fast. Building the physics-world server now means being the *default* physics tool agents reach for — a position that compounds.
- **Revenue:** usage-based API pricing (per solve / per simulation-minute), exactly the model that scales with agent adoption rather than fighting it.
- **Defensive:** if a frontier lab ever decides to "solve physics tutoring," the cheapest path is to plug into the verified world rather than rebuild 20,000 curated, machine-checked rows. You become infrastructure.

6.4 The WhisperFlow analogy, stress-tested

The founder's argument: Wispr Flow survives ChatGPT's voice mode and Gboard's autocorrect because incumbency = users + workflow integration + trust + iteration data — not raw capability. Applied here: even if a frontier model can *draw* a circuit sim, it cannot *guarantee* it, and schools/teachers/agents that depend on correctness will keep paying for the guaranteed layer. **The analogy holds, with one condition:** the trust must be *earned and demonstrated* — published verification methodology, the dimension-checker, the coverage dashboard. Trust is the moat, but only if it's visible.

7. Emergent Discovery: What's Real, What's Hype

Real — and demonstrable:

- **Novel solution paths.** The resolver enumerates *all* derivations, including ones no textbook prints (e.g., solving a circuit problem via energy bookkeeping instead of nodal analysis). New to students; sometimes new to teachers.

- **Cross-topic paths.** Because properties are topic-agnostic (§3.1), the system will chain electromagnetism → mechanics (cavity field → force → projectile) without being told the topics connect. Humans silo knowledge by chapter; the graph doesn't.
- **Gap detection.** The completeness audit (§3.3) automatically finds missing links — "12 properties in Chapter 7 have no solving formula" is the system telling you what physics it can't yet do. That's a form of self-knowledge no AI tutor has.

Hype — do not claim:

- The system will not discover new *laws*. Laws are axioms here, not outputs. Claiming otherwise invites dismissal by anyone technical.
- "The database learns by observing components" should be phrased precisely: it *derives consequences* of encoded laws combinatorially. The honest frame — *combinatorial creativity within fixed rules* — is strong enough without mysticism.

The killer demo remains: pose the hollow-cavity charged-sphere projectile problem — a scenario no sim library contains — and watch the world derive it (cavity field → qE force → acceleration → trajectory) and animate the build sequence. If it can do *that* live, the concept is proven.

8. Moat Analysis

```
Moat candidate  |  Strength  |  Notes
Curated, machine-checked tables  |  ★★★★☆  |  Compounds with every chapter; a funded clone needs 1–2 years of expert curation to match
Verification methodology (dimension-checker, audit CI)  |  ★★★★☆  |  Visible trust machinery; hard to fake, slow to replicate well
MCP default-tool position  |  ★★★☆☆  |  Winner-take-most dynamics in agent tooling; early mover advantage is real but not locked
Sequenced-animation UX  |  ★★★☆☆  |  Copyable in principle; but "every frame justified" requires the tables, so it ladders on moat #1
Brand/community (teachers, JEE students)  |  ★★★☆☆  |  Slow to build, sticky once built; India's exam-prep market is community-driven
Raw algorithms (funnel, resolver)  |  ★★☆☆☆  |  Clean but not uncopyable — correctly rated the weakest moat; don't rely on it
```

Net assessment: no single deep moat on day one; a *stack* of medium moats that compound. This is the WhisperFlow pattern — defensibility through accumulated users, trust, iteration data, and workflow embedding rather than one secret. The strategy must therefore be **velocity**: ship the narrow complete world fast, expand chapters fast, launch MCP fast, and let compounding do the rest.

9. Business Models (ranked)

- **B2C freemium (students).** Free: browse shop + limited solves/day. Paid (~₹499–999/yr India / ~$30–60/yr global — undercutting Brilliant's ~$130): unlimited solves, all paths, exam-mode problem sets. India's JEE/NEET prep market is enormous, price-sensitive, and community-driven — ideal for bottom-up adoption.
- **B2B institutional (schools/coaching).** Labster-style seat licenses with teacher dashboards, assignments, and curriculum mapping. Slower sales cycle, larger tickets. Phase 2.
- **B2D API/MCP usage.** Per-solve and per-simulation-minute pricing for AI tutors and edtech platforms. Smallest today, potentially largest by 2028 as agents proliferate.
- **Content/licensing.** License the verified tables (or verification-as-a-service) to publishers and exam boards. Long-term option.

Recommended sequencing: **1 → 3 → 2**. Consumer love first (it also generates the usage data that proves correctness at scale), agent API while the MCP window is open, institutional once the curriculum coverage is undeniable.

10. Future Scenarios

Bull case (25%)

FieldForge nails DC circuits → AC/magnetics → mechanics by end of year 2, with full JEE/NEET chapter coverage. The MCP server becomes the default physics tool for AI tutors; one major agent platform integration drives API revenue past subscriptions. A coaching giant or edtech incumbent acquires or partners. The "lawful simulation" brand becomes synonymous with trustworthy physics visuals the way PhET is synonymous with free sims.

Base case (50%)

A beloved niche product: the go-to bench for JEE/AP physics students and a respected MCP physics tool with steady API usage. Sustainable independent business on B2C + B2D revenue. Slow institutional expansion. The tables grow chapter by chapter; the moat compounds quietly.

Bear case (20%)

Curation velocity stalls (the content grind beats the founder); a funded team ships a "good enough" generative sim with a verification badge; or frontier models gain built-in formal verification that narrows the lawful-vs-plausible gap. FieldForge survives as an open-source reference implementation but not a business.

Wildcards

- **Spatial computing:** Apple Vision / XR headsets make the "virtual lab bench" literal — grabbing components with your hands. The component-world architecture is natively spatial; 2D-first competitors would need rewrites.
- **Government digital-lab pushes:** India's Virtual Labs program and NEP digital-education funding could make a verified, curriculum-mapped, free-tier world a policy-level asset — or a competitor if a state-backed alternative ships first.
- **Exam-board integration:** if a board (CBSE, etc.) blesses verified simulations for practical exams, the compliance moat is enormous.
- **AI regulation:** any future requirement that educational AI content be *verifiable* turns FieldForge's architecture from an advantage into a legal necessity for competitors.

11. Roadmap

```
Phase | Scope | Exit criteria
0 — Bench | DC circuits shop: ~40 components, nodal-analysis solver, drag-and-wire 2D canvas, probe anything | Any DC circuit builds and solves; values probeable
1 — Resolve | Problem → components → funnel → recursive resolution → multi-path derivations; Class 12 current-electricity chapter fully covered | Coverage CI: 100% of chapter's end-of-chapter problems resolve
2 — Animate | Trace → sequenced 3D build animation; schematic-first visuals | The cavity-style demo runs end to end (in its domain)
3 — Agent world | MCP server with the §6.2 tool surface; docs for agent integrators | An external AI agent builds a verified sim via MCP
4 — Cross-domain | Mechanics domain (particles + fields); EM→mechanics paths; AC + magnetics | Cavity-projectile derived and animated live
5 — Compound | Community formula workbench; chapter-by-chapter expansion; B2B pilot | Curation velocity 2× founder-only rate; first institutional customer
```

Non-goals until Phase 3: accounts/billing complexity, mobile apps, social features. Boring choices everywhere except the tables.

12. Top Risks & De-risking

```
Risk | Likelihood | Mitigation
Content curation stalls (the grind) | High | Ruthless scoping (one chapter = done); audit-driven backlog; community workbench in Phase 5; founder's EE background covers circuits natively
3D quality bar burns time/money | High | Schematic-first; renderer downstream of spec so visuals upgrade independently; 2D canvas MVP is shippable
"Paste any problem" overpromised | Medium | Builder-first UX; text = parts list, hands = wiring; never demo what the parser can't do
```

```
Funded clone  |  Medium  |  Velocity + MCP default position + trust brand; the tables are the moat and they take 1–2 years to replicate
AI verification catches up  |  Low–Med  |  Even so, *someone* must author the ground truth the verifiers check against — that someone is this project's tables
Single-founder throughput  |  High  |  Phase 0–1 are deliberately one-person-sized; raise or hire only after the cavity-style demo proves the concept
```

13. Verdict

Build it — but build it narrow first. The idea's strength is not that it covers all of physics; it is that **for any covered chapter, it is complete, verified, and explorable in ways nothing else is**. The architecture (property-hub tables, component-indexed funnel, recursive resolution, trace-as-storyboard, tools-as-components) is sound and genuinely original in combination. The market is large, growing, and geographically aligned with the founder. The MCP inversion turns the biggest threat (AI tutors) into the biggest channel. The risks are execution risks — curation velocity, visual bar, scoping discipline — which are manageable precisely because they are *known*.

The one-sentence strategy: *become the lawful physics world — first for JEE students' bench, then for every AI agent's hands — one fully-covered chapter at a time.*

Document prepared from market research conducted September 2026 (web sources: market reports, competitor sites, MCP ecosystem directories, edtech analyses) and from the founder's design sessions. Market figures are analyst estimates and should be re-validated before fundraising use. Nothing in this document is legal or financial advice.